

Forecasting Realized Volatility in Markets: Does Sentiment Add Information Beyond the HAR Framework?

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Abstract

This paper empirically investigates whether exogenous alternative data—specifically, natural language processing (NLP) derived news sentiment and market microstructure signals—contains incremental predictive power for forecasting realized volatility beyond the classical Heterogeneous Autoregressive (HAR) framework. We conduct a cross-asset evaluation using high-frequency Bitcoin (BTC), the S&P 500 (SPX), and the NIFTY 50 index. We compare a linear sentiment-augmented model (HAR-S) against a non-linear deep learning hybrid (Neural-HAR) that employs a Gated Residual Network and an FT-Transformer to predict the residual error of an ordinary least squares HAR prior. We find that linear models struggle to extract out-of-sample signal from high-dimensional sentiment data. In contrast, the Neural-HAR model yields statistically significant improvements in forecasting accuracy, primarily under the QLIKE loss function, and acts as the sole surviving model in the Model Confidence Set for SPX. However, we note an important limitation: the Neural-HAR evaluation is conducted on a shorter out-of-sample window (≈ 1 trading year) compared to the multi-year baseline HAR evaluation. In an economic application using a volatility-targeting strategy, the superior forecasts of the Neural-HAR model do not improve the Sharpe ratio, but they systematically reduce realized portfolio volatility and maximum drawdowns, offering enhanced risk control. Finally, we find no robust evidence that sentiment’s marginal predictive content is strictly regime-dependent.

Keywords: Realized volatility, HAR model, sentiment analysis, FinBERT, model confidence set, volatility forecasting, cryptocurrency

JEL: C22, C53, C58, G12, G17

1. Introduction

Modeling and forecasting realized volatility is a cornerstone of empirical finance, with direct applications in risk management, derivatives pricing, and portfolio optimization. The Heterogeneous Autoregressive (HAR) model of [1] has become the benchmark for this task due to its parsimony and its ability to capture the long-memory properties of volatility through a simple additive cascade of daily, weekly, and monthly realized variance components.

However, financial markets are increasingly driven by exogenous information flows, macroeconomic shocks, and retail sentiment that may not be fully absorbed into the autoregressive history of prices. The proliferation of natural language processing (NLP), notably models like FinBERT [2, 3], allows researchers to quantify market sentiment from news and social media at high frequencies. A burgeoning literature asks whether such alternative data adds incremental predictive power beyond the HAR baseline. Early attempts to linearly augment the HAR model with exogenous variables often suffer from the curse of dimensionality; while in-sample fit improves, out-of-sample forecasting performance deteriorates due to overfitting to noisy sentiment signals.

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This paper makes four distinct contributions to this literature. First, we conduct a rigorous cross-asset evaluation spanning a 24/7 retail-driven cryptocurrency market (Bitcoin), a deeply liquid institutional equity market (S&P 500), and an emerging market equity index (NIFTY 50). Second, we propose a hybrid architecture, Neural-HAR, which uses a linear ordinary least squares (OLS) HAR prior and a deep residual ensemble composed of a Gated Residual Network (GRN) and an FT-Transformer [4] to digest non-linear sentiment and microstructure interactions. Third, we perform a regime-conditional analysis using a Hidden Markov Model to test whether the marginal value of sentiment varies across market stress regimes. Fourth, we test the economic significance of these forecasts using a volatility-targeting framework.

Our empirical results present a nuanced picture. We find that linear sentiment-augmented models (HAR-S) generally underperform the HAR baseline out-of-sample. The non-linear Neural-HAR model, however, isolates predictive signal, outperforming the HAR baseline significantly under the QLIKE loss function and surviving in the Model Confidence Set [5]. Despite this statistical superiority, the economic significance is mixed: the Neural-HAR model does not improve risk-adjusted returns (Sharpe ratio) net of turnover costs, but it does significantly reduce maximum drawdowns and absolute realized volatility.

2. Literature Review

Our work intersects the literature on realized volatility forecasting, the econometric evaluation of predictive accuracy, and the application of deep learning to tabular financial data.

The baseline for our analysis is the HAR-RV model introduced by [1], which elegantly models the volatility cascade generated by heterogeneous market participants. While standard loss functions like Mean Squared Error (MSE) are common, [6] demonstrates that the QLIKE loss function is robust to noise in the volatility proxy, making it the preferred metric for ranking volatility forecasts. To rigorously assess forecast superiority, we rely on the Diebold-Mariano test [7] with the small-sample correction of [8], as well as the Model Confidence Set (MCS) framework of [5], which isolates a set of models that contain the best model at a given confidence level.

In recent years, the literature has increasingly explored the integration of sentiment into volatility models. The advent of domain-specific language models, such as FinBERT [2, 3], has provided researchers with robust, continuous sentiment indices. However, feeding high-dimensional, noisy sentiment data directly into linear autoregressive models often yields poor out-of-sample predictions. To address this, we draw on recent advances in deep learning for tabular data, specifically the Feature Tokenizer Transformer (FT-Transformer) and Gated Residual Networks [4]. By combining a linear prior with a deep learning residual branch, our Neural-HAR model seeks to overcome the linear constraints of classical models while avoiding the optimization instability of pure neural networks in low signal-to-noise environments.

3. Data

We analyze three structurally distinct assets: Bitcoin (BTC), the S&P 500 index (SPX), and the NIFTY 50 index (NIFTY). Table 1 presents the descriptive statistics for the daily log realized volatility of each asset. The data spans from January 23, 2019, to July 28, 2026, for BTC ($N = 2,744$), and from February 5, 2015, to July 28, 2026, for SPX ($N = 2,862$) and NIFTY ($N = 2,755$). For BTC, realized variance is computed using 5-minute intraday prices; for the equity indices, a realized volatility proxy is constructed from daily OHLCV data.

Table 1: Descriptive Statistics for Daily Log Realized Volatility

Asset	N	Start	End	Mean	Std	Min	Max	Skew	Kurt	AC(1)	AC(5)	AC(22)	LB(22)_p
BTC	2744	2019-01-23	2026-07-28	-7.396600	1.070100	-11.700100	-2.202200	0.147000	0.826000	0.702900	0.465400	0.352100	-
SPX	2862	2015-02-05	2026-07-28	-11.054300	2.591900	-27.001000	-4.116900	-1.082000	2.308000	0.117900	0.142100	0.074200	-
NIFTY50	2755	2015-02-05	2026-07-28	-10.979400	2.439100	-24.754800	-3.946000	-1.297000	3.087000	0.073300	0.062300	0.045400	-

The time series of realized volatility for all three assets visually exhibits the stylized facts of financial volatility, including volatility clustering and a pronounced, synchronous spike during the onset of the COVID-19 pandemic in early 2020.

In addition to realized volatility, our feature matrix includes a suite of exogenous variables. Market microstructure is captured via Volume-Synchronized Probability of Informed Trading (VPIN), Order Book Imbalance (OBI), and Amihud Illiquidity. Macroeconomic controls include the term slope and credit spread. Finally, sentiment is measured via a daily aggregate FinBERT score derived from news headlines, a calculated sentiment surprise metric, and retail fear/greed indices.

4. Methodology

4.1. HAR and HAR-S Models

The standard HAR model predicts the log realized volatility using daily, weekly, and monthly (22-day) lags:

$$\log(RV_t) = \alpha + \beta_d \log(RV_{t-1}) + \beta_w \log(RV_{t-5:t-1}) + \beta_m \log(RV_{t-22:t-1}) + \varepsilon_t \quad (1)$$

The Sentiment-Augmented HAR (HAR-S) linearly extends this by including the lagged exogenous feature vector X_{t-1} , which contains FinBERT sentiment, term slope, credit spread, and microstructure variables:

$$\log(RV_t) = \alpha + \beta_d \log(RV_{t-1}) + \beta_w \log(RV_{t-5:t-1}) + \beta_m \log(RV_{t-22:t-1}) + \gamma^\top X_{t-1} + \varepsilon_t \quad (2)$$

These models are estimated via Ordinary Least Squares (OLS) utilizing Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors.

4.2. The Neural-HAR Architecture

The Neural-HAR model is a hybrid architecture designed to isolate non-linear interactions without sacrificing the robust autoregressive baseline of the HAR model. The model computes a linear OLS HAR prior. In parallel, a residual neural network takes both the HAR lags and the exogenous features X_{t-1} as input to predict the residual error of the linear prior.

The residual branch is an inverse-validation-loss weighted ensemble of two deep learning architectures: 1. Gated Residual Network (GRN): Utilizes Gated Linear Units (GLU) to perform soft feature selection, dampening noisy sentiment variables while amplifying predictive signals. 2. FT-Transformer: Tokenizes numerical features and employs multi-head self-attention to model complex, higher-order cross-feature interactions.

4.3. Validation and Evaluation Protocol

To strictly prevent look-ahead bias, we employ an expanding-window walk-forward validation scheme. The baseline HAR and HAR-S models are re-estimated using all available prior data. For the Neural-HAR model, the walk-forward routine utilizes a 21-day step size with an initial minimum training window of 252 days. At each step, Optuna optimizes the hyperparameters (learning rate, weight decay, dropout, network depth) over 40 trials.

Forecasts are evaluated out-of-sample (OOS) using QLIKE, Mean Squared Error (MSE), and Mean Absolute Error (MAE).

$$\text{QLIKE}(RV_t, \widehat{RV}_t) = \frac{RV_t}{\widehat{RV}_t} - \log\left(\frac{RV_t}{\widehat{RV}_t}\right) - 1 \quad (3)$$

We apply the Diebold-Mariano (DM) test with the Harvey-Leybourne-Newbold (HLN) small-sample correction, and the Model Confidence Set (MCS) using a block bootstrap ($B = 1000$ resamples) at the $\alpha = 0.10$ significance level.

Important Limitation on Sample Alignment: Because the Neural-HAR model is computationally intensive and was evaluated over a shorter trailing period, the sample sizes for the out-of-sample evaluation differ. The baseline HAR expanding-window metrics in our preliminary descriptive statistics cover large samples ($N > 2400$). However, the direct DM and MCS comparisons involving the Neural-HAR model (and the corresponding HAR baseline for that specific evaluation) are strictly evaluated on the shortest common trailing window ($N = 275$ for BTC, $N = 287$ for SPX, $N = 276$ for NIFTY), representing approximately one recent trading year that does not include the COVID-19 pandemic shock. All relative statistical comparisons in the subsequent sections are computed on this harmonized short window.

5. Empirical Results

5.1. In-Sample Fit

Table 2 details the in-sample HAC-OLS coefficients for the HAR and HAR-S models. While the HAR-S model improves in-sample $\bar{R}^2$ modestly, the coefficients on the FinBERT sentiment score and sentiment surprise are statistically insignificant across all regimes for both SPX and NIFTY (see Section 6). For BTC the primary non-autoregressive in-sample driver is the squared order-book imbalance (OBI): $p < 0.01$ in the low- and medium-stress regimes (see Table 7); FinBERT coefficients for BTC are not identified in the HAR-S regime splits because the BTC news-sentiment series is unavailable for a sufficient in-sample window in those partitions.

5.2. Out-of-Sample Forecasting

When evaluated out-of-sample, the linear HAR-S model consistently underperforms the baseline HAR on QLIKE for BTC (DM = -1.69 , $p = 0.091$) and on both QLIKE and MSE for NIFTY (QLIKE-DM = -3.55 , $p < 0.001$). This confirms the well-known tendency for high-dimensional linear models to overfit noisy sentiment signals out-of-sample. HAR-S does not improve significantly on HAR for SPX under either loss. The full-sample OOS metrics for the baseline models are reported in Table ??; the harmonized trailing-window metrics for all three models are in Table 3. These results are sourced from `table2_dm_test.csv` (the canonical DM output of `run_paper.py`, which uses chronologically correct end-aligned array trimming).

The Neural-HAR ensemble shows a different pattern. Table 4 (sourced from `table2_dm_test.csv`) reports QLIKE improvements over the HAR baseline that are statistically significant at the 5% level for BTC (DM = $+1.98$, $p = 0.048$) and SPX (DM = $+5.05$, $p < 0.001$), and at the 5% level for NIFTY (DM = $+2.44$, $p = 0.015$). Under MSE, however, the picture reverses: HAR is preferred over Neural-HAR for all three assets (DM_{MSE} = -3.28 , -3.63 , -3.08 for BTC, SPX, NIFTY; all $p < 0.01$). This divergence between QLIKE and MSE results is consistent with the sensitivity of squared-error loss to large outlier realizations and with (author?) [6]’s argument that QLIKE is the more robust metric when the volatility proxy is itself noisy.

Note on a pipeline artifact in `table4_dm_tests.csv`. An earlier supplementary output (`table4_dm_tests.csv`, generated by `notebooks/paper_tables.py`) reported BTC Neural-HAR MSE-DM = $+3.82$ ($p < 0.001$), the opposite conclusion to the canonical result above. Source-code inspection confirms this was caused by a front-alignment bug in that notebook: it trimmed all forecast arrays from the start (`v[:min_n]`) rather than the chronological end (`v[-min_n:]`), so it was inadvertently comparing HAR forecasts from the beginning of the HAR series with Neural-HAR forecasts from the beginning of the Neural-HAR series — two non-overlapping time windows. The bug has been corrected in the notebook. Throughout this paper we rely exclusively on the output of `run_paper.py`, which has always performed correct end-alignment and whose DM results are reported in Table 4.

5.3. Model Confidence Set

Table 5 reports the Model Confidence Set (MCS) results at $\alpha = 0.10$ using QLIKE losses. For SPX, the Neural-HAR model is the sole survivor; the baseline HAR is eliminated (MCS p -value = 0.0 , `table5_mcs.csv`). For BTC and NIFTY both HAR and Neural-HAR survive the MCS, meaning the data do not provide sufficient power to eliminate either model at the 90% confidence level over this approximately one-year evaluation window. The full three-model MCS (including HAR-S) is reported in Table 6 for completeness.

6. Regime-Conditional Analysis

To investigate whether the marginal predictive content of sentiment is state-dependent, we filter the data into low, medium, and high volatility regimes using a Hidden Markov Model.

Table 7 presents the regime-conditional in-sample coefficients for the HAR-S model. Contrary to the hypothesis that sentiment is most impactful during specific market stress periods, the FinBERT sentiment coefficients are statistically insignificant across all regimes for both SPX ($p \approx 0.17 - 0.55$) and NIFTY

Table 2: In-Sample HAC-OLS Coefficients: HAR and HAR-S

Asset	Model	Variable	Coef	p-value	Sig	R^2	N
BTC	HAR	const	-1.135600	0.000000	***	0.403700	2744
BTC	HAR	RV_d	0.171400	0.000000	***	0.403700	2744
BTC	HAR	RV_w	0.375200	0.000000	***	0.403700	2744
BTC	HAR	RV_m	0.300100	0.000000	***	0.403700	2744
BTC	HAR-S	const	-1.010300	0.124000	NaN	0.187300	1095
BTC	HAR-S	RV_d	0.217500	0.000000	***	0.187300	1095
BTC	HAR-S	RV_w	0.223200	0.006000	***	0.187300	1095
BTC	HAR-S	RV_m	0.538900	0.000000	***	0.187300	1095
BTC	HAR-S	vpin_lag1	-0.794400	0.116000	NaN	0.187300	1095
BTC	HAR-S	obi_sq_lag1	2.769200	0.000000	***	0.187300	1095
BTC	HAR-S	illiq_lag1	0.000000	0.095000	*	0.187300	1095
BTC	HAR-S	FinBERT_score_lag1	0.000000	0.252000	NaN	0.187300	1095
BTC	HAR-S	sent_surprise_lag1	0.000000	0.000000	***	0.187300	1095
BTC	HAR-S	credit_spread_lag1	-0.050000	0.606000	NaN	0.187300	1095
BTC	HAR-S	term_slope_lag1	0.046200	0.645000	NaN	0.187300	1095
BTC	HAR-S	crypto_fg_lag1	0.020400	0.907000	NaN	0.187300	1095
SPX	HAR	const	-3.360000	0.000000	***	0.088500	2862
SPX	HAR	RV_d	-0.004100	0.845000	NaN	0.088500	2862
SPX	HAR	RV_w	0.248600	0.000000	***	0.088500	2862
SPX	HAR	RV_m	0.451800	0.000000	***	0.088500	2862
SPX	HAR-S	const	-9.928200	0.000000	***	0.033900	755
SPX	HAR-S	RV_d	-0.038700	0.236000	NaN	0.033900	755
SPX	HAR-S	RV_w	-0.062100	0.549000	NaN	0.033900	755
SPX	HAR-S	RV_m	0.399900	0.020000	**	0.033900	755
SPX	HAR-S	illiq_lag1	-0.000000	0.637000	NaN	0.033900	755
SPX	HAR-S	FinBERT_score_lag1	-0.235200	0.802000	NaN	0.033900	755
SPX	HAR-S	sent_surprise_lag1	-1.084200	0.222000	NaN	0.033900	755
SPX	HAR-S	credit_spread_lag1	0.628500	0.028000	**	0.033900	755
SPX	HAR-S	term_slope_lag1	0.525000	0.071000	*	0.033900	755
NIFTY50	HAR	const	-4.602000	0.000000	***	0.039900	2755
NIFTY50	HAR	RV_d	-0.016400	0.429000	NaN	0.039900	2755
NIFTY50	HAR	RV_w	0.115200	0.034000	**	0.039900	2755
NIFTY50	HAR	RV_m	0.482300	0.000000	***	0.039900	2755
NIFTY50	HAR-S	const	-10.082400	0.000000	***	0.030500	719
NIFTY50	HAR-S	RV_d	-0.006600	0.874000	NaN	0.030500	719
NIFTY50	HAR-S	RV_w	0.138000	0.172000	NaN	0.030500	719
NIFTY50	HAR-S	RV_m	0.160700	0.354000	NaN	0.030500	719
NIFTY50	HAR-S	illiq_lag1	0.000000	0.978000	NaN	0.030500	719
NIFTY50	HAR-S	FinBERT_score_lag1	-0.222700	0.803000	NaN	0.030500	719
NIFTY50	HAR-S	sent_surprise_lag1	-0.376300	0.670000	NaN	0.030500	719
NIFTY50	HAR-S	credit_spread_lag1	0.621900	0.022000	**	0.030500	719
NIFTY50	HAR-S	term_slope_lag1	0.717400	0.009000	***	0.030500	719

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. HAC standard errors.

Table 3: OOS Forecast Accuracy: All Models on Harmonised Trailing Window

Asset	Model	N_OOS	QLIKE	MSE	MAE
BTC	HAR	1119	0.676900	2.034900	1.154900
SPX	HAR	751	14.802500	7.728000	2.118300
NIFTY50	HAR	716	3.363300	6.090500	1.810200

Table 4: Diebold–Mariano Test (HLN Correction): Challenger vs. HAR Benchmark (End-Aligned Trailing Window, $N \approx 275$ –287)

Model	Asset	QLIKE_DM	QLIKE_p	QLIKE_sig	QLIKE_better	MSE_DM	MSE_p	MSE_sig	MSE_better	OOS_R2
HAR-S	BTC	-1.690300	0.091000	*	A	1.514900	0.129800	NaN	B	0.046400
Neural-HAR	BTC	1.980000	0.047700	**	B	-3.283300	0.001000	***	A	-0.152100
HAR-S	SPX	0.704100	0.481300	NaN	B	0.082100	0.934600	NaN	B	0.002200
Neural-HAR	SPX	5.053000	0.000000	***	B	-3.630500	0.000300	***	A	-0.163800
HAR-S	NIFTY	-3.551500	0.000400	***	A	-2.210700	0.027100	**	A	-0.049900
Neural-HAR	NIFTY	2.435800	0.014900	**	B	-3.077700	0.002100	***	A	-0.123500

Positive DM statistic favours the challenger model. ‘Better’ column: A = HAR preferred, B = Challenger preferred. All arrays aligned to the chronological end of the shorter series. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Model Confidence Set: Neural-HAR vs. HAR ($\alpha = 0.10$, QLIKE Loss, Trailing Window $N \approx 275$ –287)

Asset	model	mean_loss	MCS_pvalue	in_MCS	rank	eliminated_at
BTC	Neural-HAR	0.641505	1.000000	True	1	NaN
BTC	HAR	0.713312	1.000000	True	2	NaN
SPX	Neural-HAR	1.773951	1.000000	True	1	NaN
SPX	HAR	5.407347	0.000000	False	2	1.000000
NIFTY50	Neural-HAR	1.803965	1.000000	True	1	NaN
NIFTY50	HAR	2.137405	1.000000	True	2	NaN

Table 6: Model Confidence Set: All Three Models ($\alpha = 0.10$, QLIKE Loss, End-Aligned Trailing Window)

Unnamed: 0	Asset	model	mean_loss	MCS_pvalue	in_MCS	rank	eliminated_at
0	BTC	Neural-HAR	0.286585	1.000000	True	1	NaN
1	BTC	HAR	0.318403	1.000000	True	2	NaN
2	BTC	HAR-S	0.347834	1.000000	True	3	NaN
0	SPX	Neural-HAR	1.566914	1.000000	True	1	NaN
1	SPX	HAR-S	3.657654	0.000000	False	2	2.000000
2	SPX	HAR	3.896581	0.000000	False	3	1.000000
0	NIFTY	Neural-HAR	1.869842	1.000000	True	1	NaN
1	NIFTY	HAR	2.330459	0.026000	False	2	2.000000
2	NIFTY	HAR-S	3.321181	0.009000	False	3	1.000000

($p \approx 0.58 - 0.93$). Furthermore, for BTC, the order-book imbalance (OBI) metric exhibits strong significance in the low and medium regimes ($p < 0.01$), but loses statistical significance in the high-stress regime ($p = 0.12$). Regime-conditional OOS forecast accuracy is provided in Table ?? . Ultimately, we do not find robust evidence that NLP-derived sentiment’s marginal predictive content is strictly regime-dependent.

Table 7: Regime-Conditional HAC-OLS Coefficients: HAR and HAR-S

Asset	regime	model	alpha	beta_d	beta_w	beta_m	R2_in-sample	N	beta__FinBERT_score_lag1	p_FinBERT_score_lag1	beta_sent_surprise_lag1	p_sent_surprise_lag1	beta_vpin_lag1	p_vpin_lag1	beta_obi_sq_lag1	p_obi_sq_lag1
BTC	low	HAR	2.165900	0.080100	0.260700	0.381900	0.168300	447	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
BTC	low	HAR-S	-1.660100	0.271100	0.277200	0.457600	0.204100	378	0.000000	NaN	NaN	NaN	-0.699700	0.436300	3.290400	0.001800
BTC	medium	HAR	-3.526900	0.083300	0.230100	0.225300	0.094200	364	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
BTC	medium	HAR-S	-1.623900	0.221200	0.197600	0.303500	0.145900	364	0.000000	NaN	NaN	NaN	-1.031200	0.216700	3.010800	0.000900
BTC	high	HAR	-1.093900	0.203100	0.392500	0.252600	0.435000	1533	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
BTC	high	HAR-S	-0.278100	0.114600	-0.063000	0.852800	0.227400	358	0.000000	NaN	NaN	NaN	-1.208200	0.133300	1.765200	0.117700
SPX	low	HAR	-7.762400	0.061800	-0.130900	0.429500	0.006400	306	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
SPX	low	HAR-S	-10.011100	0.050300	-0.334800	0.757000	0.049600	260	0.802000	0.550900	-1.985400	0.100600	NaN	NaN	NaN	NaN
SPX	medium	HAR	-8.351400	-0.103500	0.205200	0.125900	0.014700	245	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
SPX	medium	HAR-S	-16.961200	-0.102600	0.194500	-0.288100	0.035600	245	-1.043800	0.411600	0.572100	0.567300	NaN	NaN	NaN	NaN
SPX	high	HAR	-3.057100	0.065200	0.305700	0.413800	0.114600	2311	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
SPX	high	HAR-S	-11.334900	-0.107500	-0.094200	0.394900	0.068900	250	-3.336800	0.174700	-0.175800	0.928800	NaN	NaN	NaN	NaN
NIFTY	low	HAR	-7.991200	-0.069100	0.173800	0.186600	0.013700	295	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NIFTY	low	HAR-S	-8.734600	-0.050200	0.155300	0.153200	0.020700	248	0.648200	0.583300	-0.445300	0.720000	NaN	NaN	NaN	NaN
NIFTY	medium	HAR	-6.181600	-0.041900	0.271200	0.223000	0.032800	232	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NIFTY	medium	HAR-S	-8.654000	-0.051100	0.264400	0.125900	0.056600	232	-1.202500	0.612300	-0.665700	0.695100	NaN	NaN	NaN	NaN
NIFTY	high	HAR	-4.987200	-0.067500	0.082900	0.531800	0.042300	228	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NIFTY	high	HAR-S	-14.873800	0.082500	-0.129000	0.109500	0.068400	239	0.210700	0.926700	-1.201900	0.562200	NaN	NaN	NaN	NaN

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. HAC standard errors.

7. Economic Significance

We translate our statistical forecasting accuracy into economic utility via a volatility-targeting strategy. The portfolio scales its leverage inversely to the 1-day ahead volatility forecast to target a constant risk budget (2% daily volatility), constrained by a maximum leverage of 5x. We assume a Quarter-Kelly position sizing framework and realistic transaction costs of 5 basis points (bps) per trade.

Table 8 reports the out-of-sample backtest metrics. While Neural-HAR provides statistically superior volatility forecasts (particularly for SPX), this does not translate to a higher Sharpe ratio than the baseline HAR model. The sentiment-aware Neural-HAR model exhibits significantly higher turnover (e.g., 0.365 for SPX compared to 0.093 for HAR) as it aggressively reacts to non-linear signals. Net of the 5 bps slippage, this turnover erosion degrades the risk-adjusted return.

However, from a pure risk-management perspective, the Neural-HAR model is economically valuable. By providing more accurate volatility forecasts, the Neural-HAR strategy produces lower realized annualized volatility (closer to the target mandate) and smaller maximum drawdowns for both BTC (−59.5% vs HAR’s −65.4%) and SPX (−35.2% vs HAR’s −42.3%). Thus, Neural-HAR offers better risk control, albeit at the cost of total return due to implementation friction.

Table 8: Economic Significance: Volatility-Targeting Backtest (5 bps Slippage, Quarter-Kelly, $\leq 5\times$ Leverage, $N \approx 275-287$)

Asset	Model	Sharpe	Ann_Return	Ann_Vol	MaxDD	Calmar	HitRate	Turnover	N
BTC	HAR	-1.994300	-78.550000	39.390000	-65.400000	-1.201000	0.461800	0.073700	275.000000
BTC	HAR-S	-2.031200	-85.140000	41.920000	-71.080000	-1.197800	0.461800	0.057500	275.000000
BTC	Neural-HAR	-2.095700	-70.940000	33.850000	-59.510000	-1.192200	0.461800	0.052700	275.000000
SPX	HAR	1.446200	83.370000	57.640000	-42.260000	1.972700	0.564500	0.093500	287.000000
SPX	HAR-S	1.161500	64.710000	55.720000	-50.500000	1.281500	0.550500	0.579600	287.000000
SPX	Neural-HAR	1.201100	48.770000	40.610000	-35.240000	1.384100	0.554000	0.365500	287.000000
NIFTY	HAR	-0.310700	-16.480000	53.050000	-67.280000	-0.245000	0.510900	0.050700	276.000000
NIFTY	HAR-S	-0.383100	-22.570000	58.910000	-77.110000	-0.292700	0.510900	0.165200	276.000000
NIFTY	Neural-HAR	-0.572400	-25.850000	45.160000	-64.750000	-0.399200	0.510900	0.484400	276.000000

8. Robustness and Limitations

Our findings are subject to several important limitations that warrant disclosure:

1. Evaluation Window Size: The computationally intensive nature of the Optuna-tuned walk-forward Neural-HAR model restricted its out-of-sample evaluation to approximately 275-287 trading days.

This window represents a recent, relatively calm market environment and excludes the extreme volatility clustering of the 2020 COVID-19 pandemic. The baseline HAR model’s headline statistics were computed over a much longer multi-year sample; our direct comparative tests (DM and MCS) align the samples to the shorter window, but the robustness of the neural model across rare macro shocks remains untested in this paper.

2. Regime Sample Sizes: The regime-conditional analysis partitions the data into subsets, resulting in relatively small sample sizes per regime (e.g., 155 to 260 observations for SPX and NIFTY low/medium states). This limits the statistical power of the regression coefficient tests in Table 7.
3. Implementation Costs: The economic significance tests reveal that the predictive edge of sentiment and deep learning models is highly sensitive to transaction costs. At 5 bps slippage, the alpha generated by the Neural-HAR model is fully consumed by its turnover, limiting its utility for yield-seeking portfolios, though it remains viable for downside risk control.

9. Conclusion

This paper investigated whether alternative sentiment data contains incremental predictive power for forecasting realized volatility beyond the established HAR framework. Our empirical results suggest that directly augmenting linear models with high-dimensional sentiment scores fails out-of-sample. However, when these features are processed through a non-linear deep learning architecture (Neural-HAR) comprising a Gated Residual Network and an FT-Transformer, we observe a statistically significant improvement in forecasting accuracy, primarily under the QLIKE loss function.

Despite this statistical edge and its ability to act as the sole survivor in the S&P 500 Model Confidence Set, the economic benefits of the Neural-HAR model are nuanced. The model does not generate a higher Sharpe ratio due to the transaction costs associated with its elevated turnover. Instead, its economic value lies in superior risk mitigation, specifically by yielding tighter realized volatility profiles and shallower maximum drawdowns in a volatility-targeting framework. Furthermore, we find minimal evidence supporting a strictly regime-conditional sentiment effect. Ultimately, while non-linear architectures can successfully extract volatility signals from alternative data, the practical translation of this signal into net profitability remains highly constrained by market frictions.

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Appendix A. Code and Data Availability

The full codebase, including the data pipeline, the PyTorch implementation of the Neural-HAR model, the Optuna walk-forward hyperparameter optimization routines, and the exact weights of the out-of-sample models evaluated in this study, is available in the public repository at: <https://github.com/VT69/AlphaVol-NeuralHAR-Global>.